Chapter 1. FORWARD KINEMATICS

Kinematics deals with motion i.e. position, velocity, and acceleration without considering the forces that cause the motion (Gill, 2025) and is further split into Forward and Inverse Kinematics. In this report, we focus only on Forward Kinematics.

Forward Kinematics (FK) analyses the end-effector (EE) position and orientation given a set of joint angles. This section details the FK of a 3-link manipulator with three revolute-joints, each providing 1-DoF, represented in a 2D space as shown in Fig.1.

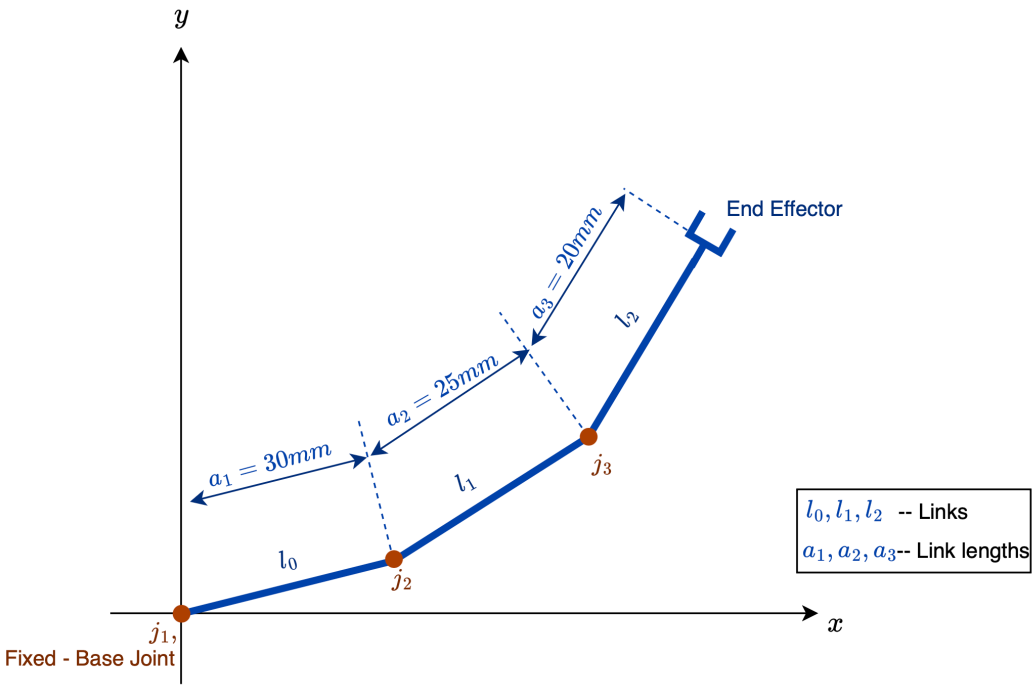

Fig.1. 3R planar manipulator

Kinematic Modeling via Denavit-Hartenberg (DH) Convention

Denavit-Hartenberg approach represents forward Kinematic chains by describing the geometry of a robot's links and joints ($e.g. n - joints$) using only a few parameters ($3n$) (Lynch,2024).

The coordinate frames are assigned by following the DH convention rules as given below (Lynch,2024; UoL,2026):

$I]x_i \perp z_{i-1}\ i.e. x_1 \perp z_0, x_2 \perp z_1, x_3 \perp z_2$

$II]x_i$ should intersect z_{i-1} – e.g. x_1 should be colinear with l_0, to intersect with z_0

$III]y_i \perp x_i\ and\ y_i \perp z_i$ in a right-handed mode

For an $n - link$ manipulator, there are $n + 1$ frames, 0 as inertial frame and $frame - n$ for EE (UoL, 2026). Hence four coordinate frames are derived in Fig.2.

Detailed steps for assigning x_i, y_i, z_i axes are given in next section.

Assigning coordinate frames

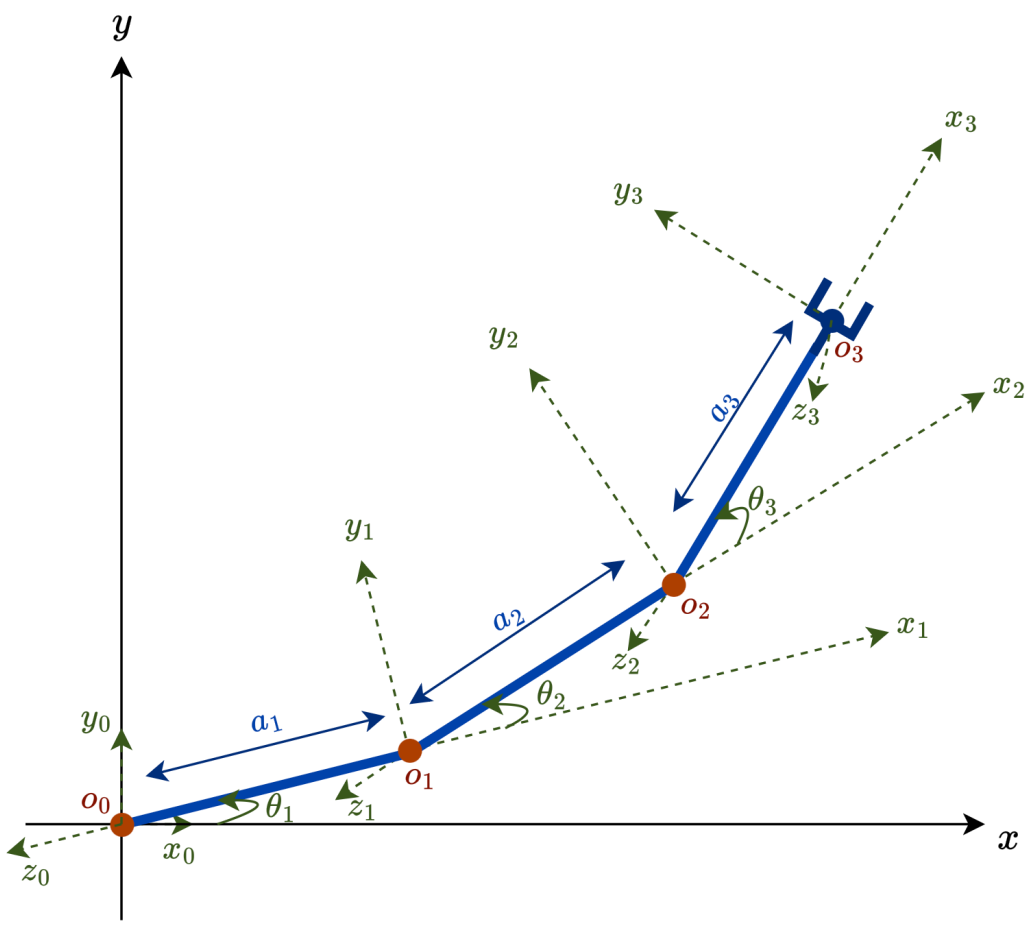

Fig. Coordinate frames

For an $n - link$ manipulator, there are $n + 1$ frames, starting with frame 0 as inertial frame and $frame - n$ for EE (UoL, 2026). Hence there are 4 coordinate frames assigned to 3-link manipulator as shown in Fig.2.

Step 1: Assigning inertial frame

Starting with inertia frame, the origin o_0 can exist at any point along z_0. x_0 and y_0 are chosen arbitrarily following the right-hand coordinate system.

Step 2: Assigning z_i axis

Regarding z_i axis, the rotation happens along the axis perpendicular to the given 2D plane. Since joint1 is revolute, z0 is the axis of rotation for joint1 and z1 for joint2. For EE joint, z3 is chosen parallel to z2, and o_n is the center of EE.

Step 3: Assigning x_i axis

As per DH convention rules mentioned earlier, x_i is chosen to be perpendicular and intersecting to z_{i-1}.

Step 3: Assigning y_i axis

y_i are not used in finding DH parameters. They are simply chosen in the direction compatible with right-hand system.

Step 4: Assigning EE coordinates:

Direction of opening and closing the gripper must be along y-axis. The x-axis and z-axis follow the intersecting principle primarily and the perpendicular rule.

DH table and Derivation of coordinate transformations

From Fig.2., the spatial relation between the base frame and EE is summarized in below DH table (UoL, 2026):

Links	θ_i (Joint angle)	d_i (Link offset)	a_i (Link length)	α_i (Link twist)
l_0	θ_1	0	a_1	0
l_1	θ_2	0	a_2	0
l_2	θ_3	0	a_3	0

Table.1. DH parameters

As per DH convention, transformation between joint i w.r.t joint $i-1$, is given by (Karhan, 2025),

$$ {}^{i-1}_{i}T = Rot(\hat{z},\theta_i)Trans(\hat{z},d_i)Trans(\hat{x},a_i)Trans(\hat{x},\alpha_i) $$

$$ = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i)\cos(\alpha_i) & \sin(\theta_i)\sin(\alpha_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i)\cos(\alpha_i) & -\cos(\theta_i)\sin(\alpha_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Substituting values from Table.1. into above equation, the individual transformation matrices A_i for each joint i, is given by,

$$ A_1 = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & a_1\cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & a_1\sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ A_2 = \begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & a_2\cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & 0 & a_2\sin(\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ A_3 = \begin{bmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & a_3\cos(\theta_3) \\ \sin(\theta_3) & \cos(\theta_3) & 0 & a_3\sin(\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

The position and orientation of the EE in inertia frame are given by the homogenous transformation (UoL, 2026),

$$ T_n^0 = A_1(q_1)A_2(q_2)\dots A_{n-1}(q_{n-1})A_n(q_n) = \begin{bmatrix} R_n^0 & o_n^0 \\ 0 & 1 \end{bmatrix} $$

$$ \therefore T_3^0 = A_1 \cdot A_2 \cdot A_3 $$

$\therefore T_3^0$

$$= \begin{bmatrix} \cos(\theta_1+\theta_2+\theta_3) & -\sin(\theta_1+\theta_2+\theta_3) & 0 & a_1\cos(\theta_1)+a_2\cos(\theta_1+\theta_2)+a_3\cos(\theta_1+\theta_2+\theta_3) \\ \sin(\theta_1+\theta_2+\theta_3) & \cos(\theta_1+\theta_2+\theta_3) & 0 & a_1\sin(\theta_1)+a_2\sin(\theta_1+\theta_2)+a_3\sin(\theta_1+\theta_2+\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Hence the coordinates of EE in cartesian space are,

$$\begin{bmatrix} x_{EE} \\ y_{EE} \\ z_{EE} \end{bmatrix} = \begin{bmatrix} a_1\cos(\theta_1)+a_2\cos(\theta_1+\theta_2)+a_3\cos(\theta_1+\theta_2+\theta_3) \\ a_1\sin(\theta_1)+a_2\sin(\theta_1+\theta_2)+a_3\sin(\theta_1+\theta_2+\theta_3) \\ 0 \end{bmatrix}$$

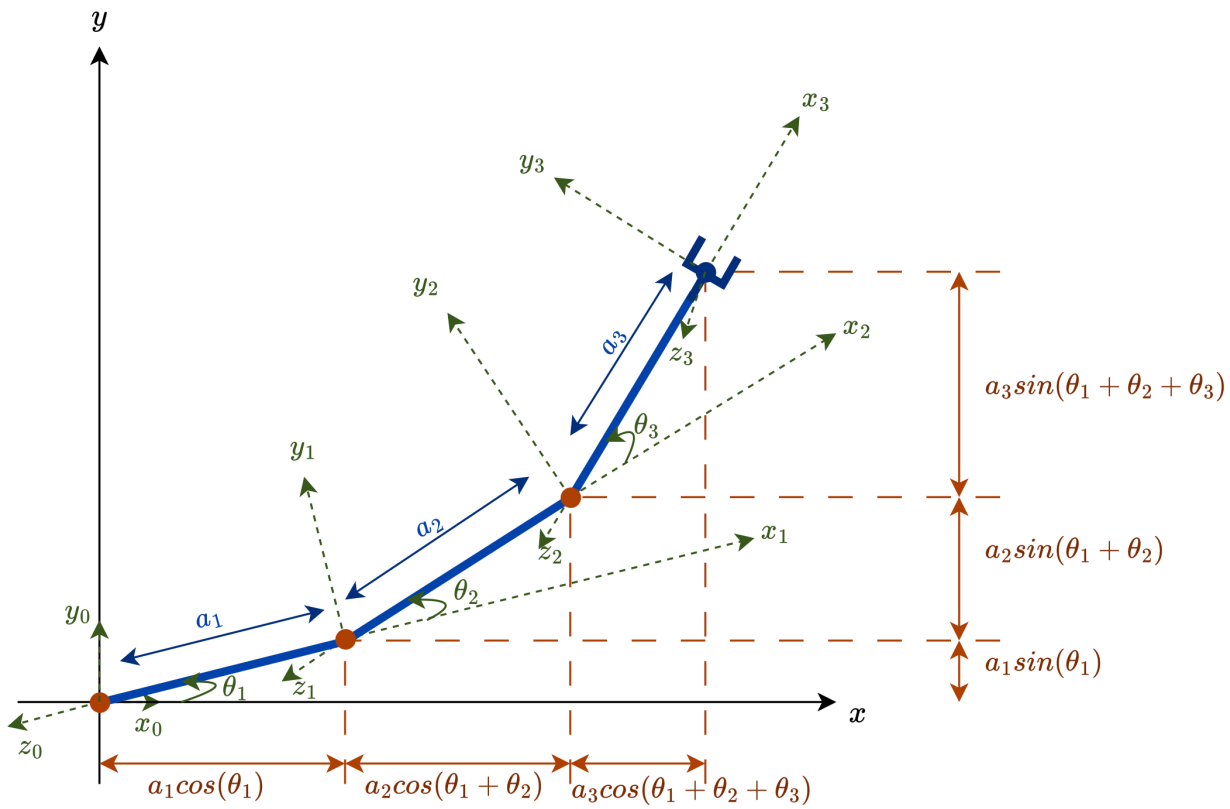

Fig.2. Kinematic schematic of 3-DoF planar manipulator

Python script Implementation:

The python script refers to these three coordinates x_{EE}, y_{EE}, z_{EE} directly in the function call forward_kinematics_calc().The user must input three angles in degrees, that represent the three joint angles, which the script internally converts into radians to calculate the (x, y) coordinates for EE position and orientation.

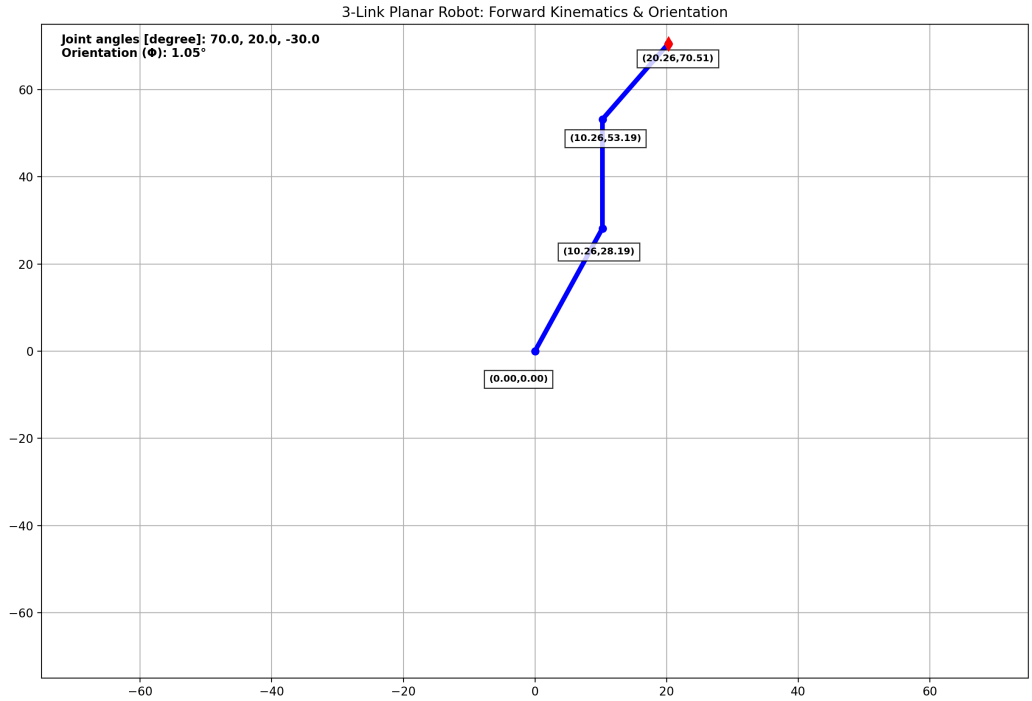

Fig.3. EE coordinates

Furthermore, the script rejects an incorrect input from user and prompts for a valid angle entry (Fig.5.). While FK model accepts arbitrary joint angles ($e.g.$ 1000°), all angular inputs here are normalized to $[0°, 180°]$ to reflect the mechanical joint limits of a 3R planar.

```
((ml_venv) ) uni@uttaras-imac project_workspace % /Users/uni/project_workspace/ml_venv/bin/python /Users/uni/project_workspace/2D_planar_robot/
MMA.py
Joint 1: Enter the Base angle in degrees: 1000
Please enter a valid numerical value for the angle between - 180 to 180 degrees.
Joint 1: Enter the Base angle in degrees: 60
Joint 2: Enter the Shoulder angle in degrees: -500
Please enter a valid numerical value for the angle between - 180 to 180 degrees.
Joint 2: Enter the Shoulder angle in degrees:
```

Fig.4. Exception Handling

Angles are calculated in radians to avoid the floating-point precision errors, e.g. $sin(90) = 1.0$ but $sin(\pi/2) = 0.89399666$. Fig.4 demonstrates the offset caused by angle in degrees during a test where a robotic link should be fully upright.

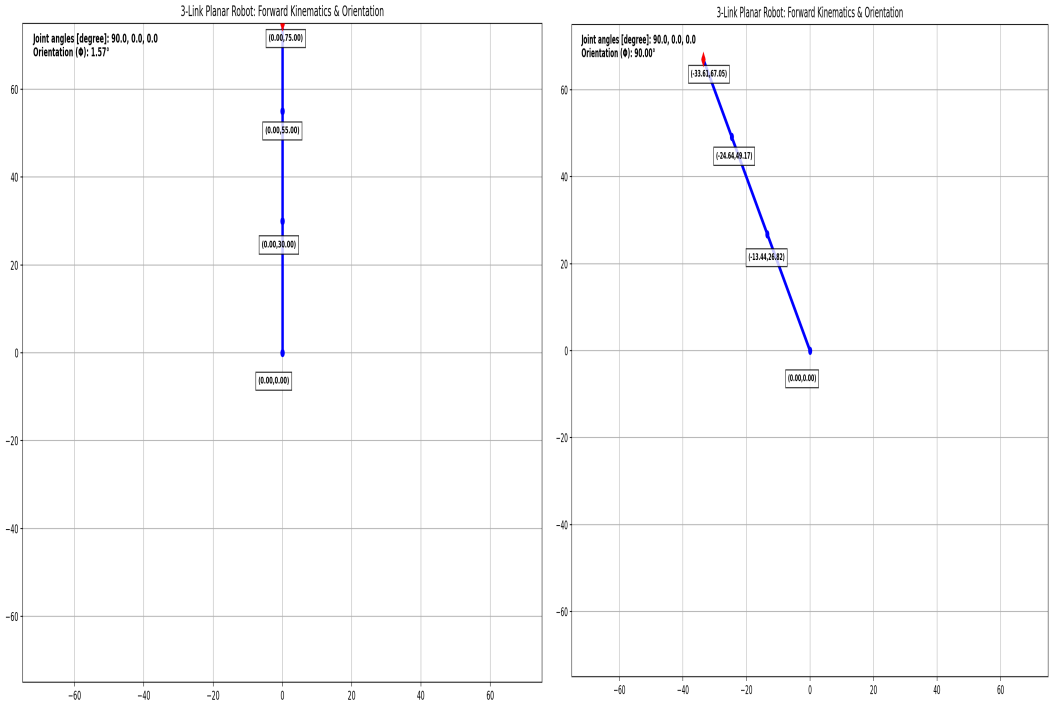

Fig.5. Vertical test with angles (90,0,0) - computed in radians (left) and in degrees (right)

REFERENCES

Gill, R. et al. (2025) Handbook of Intelligent Automation Systems Using Computer Vision and Artificial Intelligence. 1st ed. Newark: John Wiley & Sons, Incorporated.

Karhan, H. and Bingül, Z. (2025) "Automatic Determination of the Denavit–Hartenberg Parameters for the Forward Kinematics of All Serial Robots: Novel Kinematics Toolbox," Machines (Basel), 13(10), p. 944. Available at: https://doi.org/10.3390/machines13100944.

Fu, K.S., Gonzalez, R.C. and Lee, C.S.G. (1987) Robotics: control, sensing, vision, and intelligence (by) K.S. Fu, R.C. Gonzalez, C.S.G. Lee. New York: McGraw-Hill.

Lynch, K. and Park, F.C. (2024) "Appendix C Denavit–Hartenberg Parameters" in *Modern robotics : mechanics, planning, and control.* 1st ed. Cambridge, UK: Cambridge University Press. Available at: https://doi.org/10.1017/9781316661239.

UoL University of Liverpool (2026) 'Lecturecast: Robotic Manipulators', in *Week 2: Robotic Manipulators*. Available at: https://liverpool-online-study.com/course/view.php?id=3595&section=6. (Accessed on: 01 March 2026)

BIBLIOGRAPY

Cardona, M. and Serrano, F.E. (2025) "Chapter 3 Kinematics Analysis of Exoskeleton Robots" in *Rehabilitation Robotics : Kinematics, Dynamics, and Control Techniques*. 1st ed. 2025. Cham: Springer Nature Switzerland. Available at: https://doi.org/10.1007/978-3-031-83655-8.

Lynch, K. and Park, F.C. (2024) "Forward Kinematics" in *Modern robotics : mechanics, planning, and control.* 1st ed. Cambridge, UK: Cambridge University Press. Available at: https://doi.org/10.1017/9781316661239.